Introduction to the ab initio nuclear many-body problem

Lecture 1: *One- two- and many-body correlations*

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1 Introduction to many body correlations

Our aim will be to solve the many-body Schröndiger's equation from first principles, that is, without any assumptions or uncontrolled approximations. We'll work both in first and second quantization of quantum mechanics so maybe the most appropriate point of departure is establishing these two languages for the many fermion problem and the connection between them

The wavefunction of N particles obeys the many-body Schröndiger's equation

$$i\hbar\partial_t\Psi(\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_N;t) = H\Psi(\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_N;t)\ . \tag{1}$$

As a rule we'll use lower case letters (ψ, ϕ, ...) for the wavefunctions of a single particle and capital letters (Ψ, Φ, ...) wavefunctions of many particles. The many-body Hamiltonian is *many-body* operator (see below for what exactly that means) and in coordinate space it takes the form

$$H = \sum_{i=1}^{N} T(\mathbf{x}_i) + \sum_{i<j}^{N} V_2(\mathbf{x}_i,\mathbf{x}_j) + \sum_{i<j<k}^{N} V_3(\mathbf{x}_i,\mathbf{x}_j,\mathbf{x}_k) + \cdots\ , \tag{2}$$

where T is the kinetic energy operator (also to be specified clearly in a bit), and V_2, V_3, etc., are two-, three-, etc., particle potentials. Contrary to many introduction to many-particle quantum mechanics in condensed matter theory, here we explicitly allow for many-particle forces. As we will see in Lecture 3, this is an essential element in the accurate description of many-nucleon systems.

The many-body wavefunction $\Psi(\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_N)$ is a function of $d\times N$ coordinates and identifying (or inferring) its exact form is a central aim of *ab initio* nuclear theory. As with every-multivariate function we can expand the many-body wavefunction in sums of products of a complete set of functions that depend on fewer variables (as long as they provide the same asymptotic behaviours). As the simplest choice, we can choose a complete set of single-particle wavefunctions which will allow us an interpretation of each term in the sum as a configuration of particles in individual single-particle states. So we make a choice of a complete set $\{\phi_E(\mathbf{x})\}$ spanned by all possible values of E which stands for the necessary set of quantum numbers (e.g., $\mathbf{k}, s_z$ for fermions in a box) and the wavefunction becomes

$$\Psi(\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_N;t) = \sum_{E_1E_2\cdots E_N} C(E_1,E_2,\ldots,E_N;t)\phi_{E_1}(\mathbf{x}_1)\phi_{E_2}(\mathbf{x}_2)\cdots\phi_{E_N}(\mathbf{x}_N)\ . \tag{3}$$

Note that now the time-dependence is prescribed by the expansion coefficients C and that the wavefunction's normalization carries over to C as

$$\int d^dx_1d^dx_2\cdots d^dx_N|\Psi(\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_N;t)|^2 = \sum_{E_1E_2\cdots E_N}|C(E_1,E_2,\ldots,E_N)|^2 = 1 \tag{4}$$

Of course, the choice of $\{\phi_E(\mathbf{x})\}$ is far from unique and that is a good thing: any complete set of single-particle states will do, but some will do better than others. The number of terms in Eq. (3) depends heavily on the choice of the single-particle states used. A sum of many terms is harder to interpret than a sum of few, even when a single coordinate is involved, e.g., $\sum_{n=1}^{1000}(-1)^n x^{2n+1}/(2n+1)!$ has an approximate period of 2π, but that's hard to tell by inspection. Even before any attempts for interpretation, the number of terms in Eq. (3) will determine if such an approach is even practical for calculations. As such one would like to choose the set of single-particle orbitals wisely, to reduce the number of terms entering Eq. (3) and provide practical methods. Finding the expansion coefficients C would mean having the many-body wavefunction, but doing this in a brute force is hopeless. To see this explicitly we can attempt it and plug Eq. (3) into the many-body Schröndiger's equation in Eq. (1) and projecting on one set of quantum numbers(multiplying with the hermitian and integrating of all $\mathbf{x}$'s) yielding

$$
\begin{aligned}
i\hbar\partial_t C(E_1,E_2,\dots,E_N;t) &= \sum_{i=1}^{N}\sum_{W}\left[\int d^dx_i\phi^\dagger_{E_i}(x_i)T(x_i)\phi_W(x_i)\right]C(E_1,\dots,E_{i-1},W,E_{i+1},\dots,E_N;t)\\
&+\sum_{i<j}\sum_{WW'}\left[\int d^dx_i d^dx_j\phi^\dagger_{E_i}(x_i)\phi^\dagger_{E_j}(x_j)V_2(\mathbf{x}_i,\mathbf{x}_j)\phi_W(x_i)\phi_{W'}(\mathbf{x}_j)\right]C(E_1,\dots,E_{i-1},W,E_{i+1},\dots,E_{j-1},W',E_{j+1},\dots,E_N;t)\\
&+\sum_{i<j<k}\sum_{WW'W''}\left[\int d^dx_i d^dx_j d^dx_k\phi^\dagger_{E_i}(x_i)\phi^\dagger_{E_j}(x_j)\phi^\dagger_{E_k}(x_k)V_3(\mathbf{x}_i,\mathbf{x}_j)\phi_W(x_i)\phi_{W'}(\mathbf{x}_j)\phi_{W'}(\mathbf{x}_k)\right]\times\\
&\quad\times C(E_1,\dots,E_{i-1},W,E_{i+1},\dots,E_{j-1},W',E_{j+1},\dots,E_{k-1},W'',E_{k+1},\dots,E_N;t)\ . \qquad (5)
\end{aligned}
$$

For each set of quantum numbers $\{E_1,\dots,E_N\}$, Eq. (5) yields a set of differential equations. Even when truncated, this approach is hopeless for most practical cases.

Note that up to this point we haven't mentioned anything about the particles' statistics, that is, the terms in the sum in Eq. (3) have some symmetries stemming from the indistinguishability of quantum particles. For fermions this manifests as a property of the wavefunction

$$\Psi(\mathbf{x}_1,\mathbf{x}_2,\dots,\mathbf{x}_i,\dots,\mathbf{x}_j,\mathbf{x}_N;t) = -\Psi(\mathbf{x}_1,\mathbf{x}_2,\dots,\mathbf{x}_j,\dots,\mathbf{x}_i,\mathbf{x}_N;t)\ , \qquad (6)$$

where for bosons RHS's sign would be positive. This property transfers to the expansion coefficients C as

$$C(E_1,E_2,\dots,E_i,\dots,E_j,E_N;t) = -C(E_1,E_2,\dots,E_j,\dots,E_i,E_N;t)\ . \qquad (7)$$

From this the Pauli exclusion principle follows which states that if two quantum numbers are the same, e.g., $E_1 = E_2$, then C vanishes. That is, in each configuration that describes the many-body wavefunction, no two particles can be in the same state. As we'll see later this introduces the lowest rank of correlations to the system because each particle knows some information about what other particles are doing. More importantly the indistinguishability of particles also allows for a more convenient description in terms of occupation numbers, that is, we can describe each configuration by the number of particles that occupy each state in the set $\{E\}$. Forgetting for the moment that only one fermion at a time can occupy state each E, we can define the coefficients

$$\bar{C}(n_1,n_2,\dots,n_\infty) = C(E_1,E_1,\underset{n_1}{\cdots},E_2,E_2,\underset{n_2}{\cdots}\cdots,E_2,\dots)\ , \qquad (8)$$

Note that a single permutation of the arguments of C on the LHS would just introduce a minus sign. We could now parse all terms in Eq. (3) and replace all instances of $C(E_1,E_1,\underset{n_1}{\cdots},E_2,E_2,\underset{n_2}{\cdots}\cdots,E_2,\dots)$ and its permutations multiplied by the appropriate minus signs. We would be making $N!/n_1!n_2!\cdots n_\infty!$ replacements of a C coefficient with the *same* $\bar{C}$ coefficient times a sign. This means that in Eq. (4), every $N!/n_1!n_2!\cdots n_\infty!$ terms will be the same:

$$1 = \sum_{E_1E_2\cdots E_N}|C(E_1,E_2,\dots,E_N)|^2 = \sum_{n_1n_2\cdots n_\infty}|\bar{C}(n_1,n_2,\dots,n_\infty)|^2\frac{N!}{n_1!n_2!\cdots n_\infty!} \qquad (9)$$

This tells of a symmetry which we have already specified: the wavefunction is fully antisymmetric under the exchange of particles. To exploit this symmetry we define two things. First, we define the coefficients f

$$f(n_1,n_2,\dots,n_\infty) = \left(\frac{N!}{n_1!n_2!\cdots n_\infty!}\right)^{\frac{1}{2}}\bar{C}(n_1,n_2,\dots,n_\infty) \qquad (10)$$

such that $\sum_{n_1\cdots n_\infty}|f(n_1,\dots,n_\infty|^2 = 1$. Second, we examine each replacement we made realizing that a minus sign was introduced only when the C coefficient that was replaced was an odd permutation of particle indices away from the one in Eq. (8). This means that if we group all terms that correspond to a set of occupation numbers $n_1, n_2, \ldots, n_\infty$, we can define a many-particle wavefunction

$$\Phi_{n_1 n_2 \cdots n_\infty}(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N) = \left(\frac{n_1! n_2! \cdots n_\infty!}{N!}\right)^{\frac{1}{2}} \sum_{\sigma \in \mathcal{P}_N} (-1)^\sigma \phi_{E_1}(\mathbf{x}_\sigma(1)) \phi_{E_2}(\mathbf{x}_\sigma(2)) \cdots \phi_{E_N}(\mathbf{x}_\sigma(N)) \tag{11}$$

$$= \left(\frac{n_1! n_2! \cdots n_\infty!}{N!}\right)^{\frac{1}{2}} \begin{vmatrix} \phi_{E_1}(\mathbf{x}_1) & \phi_{E_1}(\mathbf{x}_2) & \cdots & \phi_{E_1}(\mathbf{x}_N) \\ \phi_{E_2}(\mathbf{x}_1) & \phi_{E_2}(\mathbf{x}_2) & \cdots & \phi_{E_2}(\mathbf{x}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{E_N}(\mathbf{x}_1) & \phi_{E_N}(\mathbf{x}_2) & \cdots & \phi_{E_N}(\mathbf{x}_N) \end{vmatrix} , \tag{12}$$

where we have introduced the inverse of the combinatorics factor we discussed before for later convenience, σ is any permutation in the permutation group $\mathcal{P}_N$ that contains all permutations of N objects and $(-1)^\sigma$ is positive (negative) for even (odd) permutations. For the second line of Eq. (12), we recognized the definition of a determinant. This object is called a *Slater determinant* and we'll list its properties further below. For now we'll only note that they are a complete basis of antisymmetric functions of many-variables. Putting all definitions together, we can finally write Eq. (3) in terms of the occupation numbers,

$$\Psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N) = \sum_{n_1 n_2 \cdots n_\infty}^{1} f(n_1, n_2, \ldots, n_\infty) \Phi_{n_1, n_2, \ldots, n_\infty}(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N) \ . \tag{13}$$

This is the form of the wavefunction in second quantization where we use occupation numbers instead of sets of quantum numbers and each term in the expansion is explicitly antisymmetric.

In order to take full advantage of the simplification that second quantization has to offer, one also defines the state vector

$$|n_1 n_2 \cdots n_\infty\rangle \tag{14}$$

such that

$$\langle \mathbf{x}_1 \mathbf{x}_2 \cdots \mathbf{x}_N | n_1 n_2 \cdots n_\infty \rangle = \Phi_{n_1 n_2 \cdots n_\infty}(\mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_N) \tag{15}$$

and corresponding creation and annihilation operators, $a_i, a_i^\dagger$ such that

$$a_i^\dagger |n_1 n_2 \cdots n_i \cdots n_\infty\rangle = \sqrt{n_i + 1} |n_1 n_2 \cdots n_i + 1 \cdots n_\infty\rangle \ , \tag{16}$$

$$a_i |n_1 n_2 \cdots n_i \cdots n_\infty\rangle = \sqrt{n_i} |n_1 n_2 \cdots n_i - 1 \cdots n_\infty\rangle \ , \tag{17}$$

which obey the *anti-commutation* relations

$$\{a_i, a_j\} = 0 \tag{18}$$

$$\{a_i, a_j^\dagger\} = \delta_{ij} \ . \tag{19}$$

Note that this means that the creation and annihilation operators carry with them (in their algebra) the commutation relation such that one doesn't have to worry about them. Further, $\langle \mathbf{x} | a_i^\dagger |0\rangle = \phi_i(\mathbf{x})$, that is, each creation or annihilation operator carries with it a single-particle wavefunction. One then finds that each state vector can be constructed as

$$\left(a_1^\dagger\right)^{n_1} \left(a_2^\dagger\right)^{n_2} \cdots \left(a_\infty^\dagger\right)^{n_\infty} |0\rangle = \prod_{i=1}^{\infty} \left(a_i^\dagger\right)^{n_i} |0\rangle = |n_1 n_2 \cdots n_\infty\rangle \ , \tag{20}$$

where

$$|0\rangle = |00\cdots 0\rangle \tag{21}$$

is the vacuum state. Putting all these definitions together and with some manipulation one arrives at the second quantized version of the Hamiltonian

$$H = \sum_{ij} \langle i|\, T\, |j\rangle\, a_i^\dagger a_j + \frac{1}{2}\sum_{ijkl} \langle ij|\, V_2\, |kl\rangle\, a_i^\dagger a_j^\dagger a_l a_k + \frac{1}{3!}\sum_{ijklmn} \langle ijk|\, V_3\, |lmn\rangle\, a_i^\dagger a_j^\dagger a_k^\dagger a_n a_m a_l + \cdots \tag{22}$$

where note the reverse ordering of the annihilation operators. We have come full circle back to the many-body Hamiltonian but now in second quantization. This definition can be extended now to any other operator $n-body$ operator A

$$A = \sum_{i_1 \neq i_2 \neq \cdots i_n = 1}^{N} A(\mathbf{x}_{i_1}, \mathbf{x}_{i_2}, \ldots, \mathbf{x}_{i_n}) = \sum_{\substack{k_1 k_2 \cdots k_n \\ l_1 l_2 \cdots l_n}} \langle k_1 k_2 \cdots k_n|\, A\, |l_1 l_2 \cdots l_n\rangle\, a_{k_1}^\dagger a_{k_2}^\dagger \cdots a_{k_n}^\dagger a_{l_n} \cdots a_{l_2} a_{l_1}\ , \tag{23}$$

where now the definition of a n-body operator is clear: when acting on a state $|n_1 n_2 \cdots n_\infty\rangle$ it redistributes n occupation numbers.

1.0.1 The effects of antisymmetrization

2 The one- and two-body problem

2.1 The one-body problem

2.2 The two-body problem

3 The many-body problem: Mean-field and Slater Determinants

3.1 The many-body problem and antisymmetrization

3.2 Mean-field description

4 The many body problem: Cooper Instability and hint of BCS determinants

4.1 Cooper's instability

4.2 BCS determinants